

# HPLC (GP) — Mean Final Regret ± SEM (n=20 variants × 3 runs)

|             | Random     | EI         | UCB        | PI         | FunBO      | PFNs4BO    | TAF_me     | TAF_ranking | TuRBO           | CAP-PPO         |
|-------------|------------|------------|------------|------------|------------|------------|------------|-------------|-----------------|-----------------|
| <b>0.5</b>  | 392.0±23.4 | 239.9±19.0 | 259.0±21.4 | 213.7±21.5 | 250.8±19.6 | 92.6±6.9   | 116.1±6.5  | 109.4±7.1   | 73.1±10.3       | <b>59.4±7.9</b> |
| <b>0.75</b> | 380.1±21.2 | 241.4±24.2 | 206.9±16.5 | 196.9±21.2 | 249.3±16.5 | 121.4±11.0 | 116.6±7.7  | 111.4±8.5   | 59.2±9.6        | <b>52.6±7.3</b> |
| <b>1.0</b>  | 356.9±14.5 | 238.3±25.0 | 207.5±15.3 | 186.5±20.0 | 206.4±15.5 | 114.5±8.4  | 111.4±8.0  | 113.4±10.9  | <b>45.5±6.3</b> | 66.5±7.3        |
| <b>1.25</b> | 411.9±20.9 | 216.6±12.1 | 259.1±16.9 | 194.4±17.3 | 253.2±17.6 | 115.6±9.1  | 143.6±8.4  | 116.0±10.6  | <b>59.7±7.6</b> | 69.7±7.5        |
| <b>1.5</b>  | 380.5±20.7 | 185.6±16.2 | 211.9±15.3 | 181.4±16.2 | 210.1±13.1 | 105.0±8.8  | 153.5±20.3 | 139.6±13.5  | <b>58.3±8.4</b> | 72.6±8.3        |
| <b>1.75</b> | 342.7±20.7 | 202.0±21.6 | 213.7±15.1 | 184.6±13.5 | 210.5±18.1 | 120.2±9.2  | 145.6±12.5 | 130.3±5.7   | <b>56.8±5.0</b> | 76.3±10.9       |
| <b>2.0</b>  | 348.0±17.9 | 182.5±14.8 | 242.2±20.3 | 202.6±18.5 | 200.8±14.8 | 115.9±8.3  | 162.3±17.9 | 138.2±17.2  | <b>50.6±6.3</b> | 72.8±6.8        |
| <b>Avg</b>  | 373.2      | 215.2      | 228.6      | 194.3      | 225.9      | 112.2      | 135.6      | 122.6       | <b>57.6</b>     | 67.1            |